

Programming for Essential Digital Skills, Chapters 1-2

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What is Python?

- Python is a *programming language*.
- A programming language is a way to instruct a computer to perform operations via written text.
- When programming, we need to be very exact. Otherwise the computer will throw an error (*syntax error*) or perhaps even do something we didn't intend it to do (*semantic error*)!
- To code Python we will use *Spyder*. This is a desktop application where you can write Python code and execute Python programs.

Why Learn Python?

Python has many benefits over some alternatives:

- It is free and open source.
- It has a large active community creating packages and providing support.
- Easier to learn than some alternatives.
- It is extremely versatile. For example, the online book was made with Python!
- It is also the most common language used in data science.

Installing Python

- To download and install Python and Spyder, go to <https://www.anaconda.com/download>.
- Follow all the default settings.
- Afterwards, open Spyder.